

Feature-Level Anomaly Generation Experiments

Experiment Plan, Hypotheses, and Interpretation

SimpleNet feature-anomaly research project

Abstract

This document explains the experiment suite implemented for the SimpleNet feature-anomaly project. The purpose is to test whether synthetic anomalies at the feature level can be made more useful than SimpleNet’s original random Gaussian noise. Each experiment changes one controlled aspect of the anomaly generator: anchor, radius calibration, spatial sparsity, or discriminator-guided refinement. The intended output is not only an AUROC table, but a causal story about which geometric property of feature-space anomalies matters for image detection and pixel localization.

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1 Research Question

SimpleNet trains a discriminator using normal patch features as positives and synthetic anomalous features as negatives. In the original setup, fake features are produced by adding isotropic Gaussian noise:

$$\mathbf{x}_{\text{fake}} = \mathbf{x}_{\text{real}} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 I). \quad (1)$$

This is simple and fast, but it ignores the geometry of the normal feature distribution. The central hypothesis of this project is:

Feature-level anomalies are more useful when they are anchored on real normal features, move in meaningful local feature directions, have calibrated magnitude, and provide spatially localized supervision.

The experiments therefore compare random noise against a geometry-aware family of anomalies derived from the fitted per-patch model

$$\Sigma_p = \mathbf{U}_p \boldsymbol{\Lambda}_p \mathbf{U}_p^\top + \varepsilon_p I, \quad (2)$$

where p indexes spatial patch location. In the most important setting, an anomaly is generated as

$$\mathbf{x}_{\text{fake},p} = \mathbf{x}_{\text{real},p} + r_p \mathbf{U}_p \sqrt{\boldsymbol{\Lambda}_p} \mathbf{v}, \quad \mathbf{v} \sim \text{Uniform}(\mathbb{S}^{k-1}). \quad (3)$$

2 Provenance of the Ideas

This section records where each idea in the experiment suite comes from. The goal is not to claim that every component is novel in isolation. The goal is to make clear which prior ideas are being combined and what new question this project asks.

2.1 SimpleNet: the Baseline to Improve

The starting point is SimpleNet [1]. SimpleNet introduced the specific training pattern used in this project: extract local features from a pretrained backbone, optionally adapt them, synthesize anomalous features by adding Gaussian noise to normal features, and train a binary discriminator. This project keeps the SimpleNet discriminator framing but questions whether the feature-level anomaly generator should be isotropic random noise.

Taken idea. Feature-space synthetic negatives can train a discriminator for anomaly detection.

Project modification. Replace directionless Gaussian noise with geometry-aware feature perturbations.

2.2 PaDiM: Patch-Wise Gaussian Feature Geometry

The use of a per-patch Gaussian model is inspired by PaDiM [2]. PaDiM models normal patch embeddings with multivariate Gaussian distributions and uses Mahalanobis distance for anomaly scoring. Our project borrows the patch-wise normal-distribution view, but uses it to synthesize training negatives rather than only to score test samples.

Taken idea. Normal patch features have spatially specific distributions that can be modeled with Gaussian statistics.

Project modification. Use the fitted covariance directions $\mathbf{U}_p$ and scales $\mathbf{\Lambda}_p$ to generate fake features for discriminator training.

2.3 PatchCore: Respecting the Normal Feature Manifold

PatchCore [3] motivates the broader view that nominal patch features themselves are the important object. It uses a representative memory bank of normal patch features and nearest-neighbor distance at inference. This project does not use a memory bank for scoring, but the anchored generator is conceptually aligned with PatchCore’s emphasis on real normal patch features: fake samples should start from real normal anchors rather than from an abstract mean alone.

Taken idea. Real normal patch features provide a strong empirical representation of the normal manifold.

Project modification. Anchor synthetic negatives on real training features:

$$\mathbf{x}_{\text{fake},p} = \mathbf{x}_{\text{real},p} + \Delta_p.$$

2.4 CutPaste, NSA, DRAEM, and DeSTSeg: Synthetic Local Supervision

Image-level anomaly synthesis methods motivate the sparse-mask experiments. CutPaste [4] uses a simple cut-and-paste augmentation as a self-supervised anomaly signal. NSA [5] creates more natural synthetic anomalies by blending patches. DRAEM [6] and DeSTSeg [7] show that synthetic corruptions and their masks can provide direct localization supervision.

Taken idea. Synthetic anomalies are more useful for localization when they are spatially local and come with an implicit or explicit mask.

Project modification. Apply synthetic perturbations only to selected feature patches and compute fake loss only on those patches:

$$\mathcal{L}_{\text{fake}} = \frac{1}{|\Omega|} \sum_{p \in \Omega} \max(0, D(\mathbf{x}_{\text{fake},p}) + m).$$

2.5 GLASS: Gradient-Guided Hard Feature Negatives

GLASS [8] is the closest recent inspiration for the gradient-refinement experiments. It argues that anomaly synthesis should be controllable and should cover weak, near-in-distribution defects. Its feature-level synthesis uses gradient ascent and projection constraints to create useful hard negatives.

Taken idea. Feature-level synthetic anomalies should not be purely random; they can be guided by the current model to produce harder, more informative negatives.

Project modification. Start from anchored PCA anomalies, then take discriminator-guided refinement steps while projecting updates back into the fitted patch PCA subspace and clamping their Mahalanobis radius.

2.6 CRAS: Distance-Guided Magnitude Calibration

CRAS [9] motivates the radius-calibration experiments. Its anomaly synthesis strategy adaptively adjusts noise based on normal-data distribution to reduce normal/anomaly overlap. This supports our suspicion that a single global noise scale, or a raw Mahalanobis threshold radius, may be poorly calibrated for training.

Taken idea. Synthetic anomaly magnitude should depend on the position of the normal sample relative to the normal distribution.

Project modification. Introduce `patch` and `anchor` radius modes:

$$r_p = \rho \sqrt{T_p / CU(0, 1)}, \quad r_{i,p} = \rho \max \left(\sqrt{T_p} - \sqrt{s(\mathbf{x}_{i,p})}, 0.05 \sqrt{T_p} \right) U(0, 1).$$

2.7 Summary of Influence

Project component	Main source idea	What we changed
Vanilla noise baseline	SimpleNet	Re-run in current controlled path
Patch Gaussian/PCA model	PaDiM	Use for synthesis, not only scoring
Real-feature anchor	PatchCore	Perturb anchors rather than only score them
Sparse patch masks	Image-level anomaly synthesis	Move masks to feature space
Gradient refinement	GLASS	Constrain updates to per-patch PCA geometry
Radius calibration	CRAS	Use patch/anchor distance in Mahalanobis units

3 Common Experimental Setup

3.1 Single-Class Protocol

The scripts are designed to run one MVTec class at a time. This reduces iteration time and makes failure analysis easier. The class is selected with `CLASSNAME`; if unset, the default is `screw`.

```
CLASSNAME=screw bash experiments/run_01_anchored_threshold.sh
```

Common run-time overrides are:

```
CLASSNAME=screw
DATAPATH=/path/to/mvtec
GPU=0
SEED=0
META_EPOCHS=40
GAN_EPOCHS=4
BATCH_SIZE=8
RESULTS_PATH=results
```

3.2 Metrics

The primary metrics are:

Image AUROC

Measures whether the detector can rank defective images above normal images.

Pixel AUROC Measures whether high anomaly scores land on anomalous pixels.

PRO Region-level localization metric; useful when defects are spatial objects rather than isolated pixels.

Anomaly-pixel AUROC

Measures localization quality inside defective images. This is especially important because the current project state suggests image-level detection is ahead of pixel-level precision.

3.3 Training Diagnostics

The discriminator logs:

<code>p_true</code>	Fraction of real normal features classified past the positive margin.
<code>p_fake</code>	Fraction of synthetic fake features classified past the negative margin.
<code>fake_patch_ratio</code>	Fraction of patches marked synthetic in sparse-mask experiments.
<code>loss</code>	Margin loss used to train the discriminator.

The important warning is that a high `p_fake` alone is not success. If the fake samples are too easy or too far from realistic defects, the discriminator can learn a boundary that improves training loss while hurting localization.

4 Experiment Ladder

4.1 Experiment 0: Vanilla SimpleNet Noise

Script.

```
bash experiments/run_00_simplenet_noise.sh
```

Configuration.

```
TSLRG_ANOMALY_MODE=simplenet_noise
TSLRG_PATCH_MASK_MODE=all
TSLRG_REFINE_STEPS=0
```

What it tests. This is the clean baseline. It restores SimpleNet-style feature noise inside the same current training path [1]:

$$\mathbf{x}_{\text{fake}} = \mathbf{x}_{\text{real}} + \mathcal{N}(0, \sigma^2 I). \quad (4)$$

Reasoning. Any proposed feature generator must beat this baseline under the same backbone, projection, discriminator, loss, dataset split, and evaluation code. Without this baseline, improvements may come from incidental training differences rather than from the anomaly generator.

Expected behavior. This baseline should be stable and may localize surprisingly well because every patch is fake and the perturbation is small. Its weakness is geometric: noise is spent equally in every channel direction, including directions where normal features have little or no meaningful variation.

Interpretation. If geometry-aware variants do not beat this baseline, then either the proposed feature geometry is not aligned with real defects, or the synthetic perturbation magnitude is miscalibrated. This experiment is the anchor for all claims.

4.2 Experiment 1: Anchored PCA, Threshold Radius

Script.

```
bash experiments/run_01_anchored_threshold.sh
```

Configuration.

```
TSLRG_ANOMALY_MODE=anchored
TSLRG_RADIUS_MODE=threshold
TSLRG_PATCH_MASK_MODE=all
TSLRG_REFINE_STEPS=0
```

What it tests. This is the previous best geometry-aware setting. Each fake feature is anchored on a real normal feature and shifted inside the fitted patch PCA subspace:

$$\mathbf{x}_{\text{fake},p} = \mathbf{x}_{\text{real},p} + r_p \mathbf{U}_p \sqrt{\mathbf{\Lambda}_p} \mathbf{v}. \quad (5)$$

The radius still follows the fitted Mahalanobis threshold:

$$r_p = \sqrt{T_p} + \delta U(0, 1). \quad (6)$$

Reasoning. Anchoring removes one layer of modeling error: the base point is a real feature, not a Gaussian draw. The PCA direction avoids wasting perturbation budget in directions unsupported by normal data.

Expected behavior. This setting should preserve the earlier image-level gain. However, because $\sqrt{T_p}$ can be around 40 in a 1536-dimensional feature space, the shift may be much larger than subtle real defects. That can explain good image AUROC but weak anomaly-pixel AUROC.

Interpretation. If image AUROC is strong but localization is weak, the issue is likely not the anchor or subspace direction alone. The next suspect is radius magnitude.

4.3 Experiment 2: Anchored PCA, Fixed Small Radius

Script.

```
bash experiments/run_02_anchored_fixed_radius.sh
```

Configuration.

```
TSLRG_ANOMALY_MODE=anchored
TSLRG_RADIUS_MODE=threshold
TSLRG_RADIUS=1
TSLRG_PATCH_MASK_MODE=all
TSLRG_REFINE_STEPS=0
```

What it tests. This decouples anomaly magnitude from the fitted threshold:

$$r_p = \rho U(0, 1), \tag{7}$$

where $\rho = \text{TSLRG_RADIUS}$.

Reasoning. The statistical boundary T_p answers a density-model question: how far is the edge of the normal cloud in Mahalanobis units? But discriminator training may need a different magnitude: small enough to mimic subtle defects, large enough to be learnable.

Expected behavior. A fixed small radius may improve localization if the threshold-radius anomalies were too extreme. It may reduce image AUROC if the anomalies become too subtle for early discriminator training.

Interpretation. If this improves pixel metrics but hurts image AUROC, the generator needs a curriculum or hard-negative schedule. If both improve, threshold radius was too large for the task.

4.4 Experiment 3: Patch-Calibrated Radius Sweep

Script.

```
bash experiments/run_03_patch_radius_sweep.sh
```

Default sweep.

```
RADII="0.25 0.5 1 2 5"
```

Configuration.

```
TSLRG_ANOMALY_MODE=anchored
TSLRG_RADIUS_MODE=patch
TSLRG_PATCH_MASK_MODE=all
TSLRG_REFINE_STEPS=0
```

What it tests. The radius is normalized by the patch’s fitted threshold and feature dimension. This combines the patch-distribution view from PaDiM [2] with the distance-calibration motivation of CRAS [9]:

$$r_p = \rho \sqrt{\frac{T_p}{C}} U(0, 1). \quad (8)$$

Reasoning. Different patches can have different thresholds because their normal feature variation differs. A global radius may under-perturb high-variance patches and over-perturb low-variance patches. Patch normalization makes a radius sweep more comparable across spatial locations.

Expected behavior. This should be more stable across classes and patches than a fixed radius. It is especially useful if per-patch T_p varies substantially.

Interpretation. If patch calibration beats fixed radius, then local scale matters. If it does not, then the dominant issue may be direction distribution or sparse supervision rather than per-patch magnitude.

4.5 Experiment 4: Anchor-Gap Radius Sweep

Script.

```
bash experiments/run_04_anchor_radius_sweep.sh
```

Default sweep.

```
RADII="0.25 0.5 1 2"
```

Configuration.

```
TSLRG_ANOMALY_MODE=anchored
TSLRG_RADIUS_MODE=anchor
TSLRG_PATCH_MASK_MODE=all
TSLRG_REFINE_STEPS=0
```

What it tests. The radius depends on how close the real anchor already is to the fitted threshold, following the same high-level principle as distance-guided anomaly synthesis in CRAS [9]:

$$r_{i,p} = \rho \max \left(\sqrt{T_p} - \sqrt{s(\mathbf{x}_{i,p})}, 0.05\sqrt{T_p} \right) U(0, 1), \quad (9)$$

where $s(\cdot)$ is the fitted Mahalanobis squared score.

Reasoning. Not all normal anchors are equally central. A feature already near the edge of the normal distribution should require a smaller shift to become a useful near-boundary negative. A very central feature can tolerate a larger shift.

Expected behavior. This should create more uniformly hard negatives: not trivially fake, but not identical to normal features. It is the most direct test of the “near-boundary negative” hypothesis.

Interpretation. If anchor-gap radius improves localization, the project has evidence that negative sample difficulty matters. If it only improves image AUROC, sparse patch supervision may still be needed.

4.6 Experiment 5: Sparse Random Patch Sweep

Script.

```
bash experiments/run_05_sparse_random_sweep.sh
```

Default sweep.

```
RATIOS="0.02 0.05 0.10 0.20"
```

Configuration.

```
TSLRG_ANOMALY_MODE=anchored
TSLRG_RADIUS_MODE=anchor
TSLRG_RADIUS=1
TSLRG_PATCH_MASK_MODE=random
```

What it tests. Only a random subset of patches is synthetic. Unselected patches are kept equal to their real anchors, and the fake loss is applied only to selected patches.

Reasoning. Real industrial defects are usually sparse. Image-level synthetic anomaly methods such as CutPaste [4], NSA [5], DRAEM [6], and DeSTSeg [7] motivate using localized synthetic supervision. If every patch in every fake image is synthetic, the discriminator may learn image-level distribution shifts rather than local defect evidence. Random sparse masks force the discriminator to distinguish local deviations in the context of mostly normal features.

Expected behavior. Sparse random patches may improve pixel AUROC and PRO. Very small ratios may under-train the fake class; very large ratios approach the all-fake setting.

Interpretation. If sparse random masks improve localization, the previous bottleneck was not only feature magnitude; it was also the spatial training signal. If image AUROC drops sharply, the fake signal may be too rare and needs a ratio curriculum.

4.7 Experiment 6: Sparse Block Patch Sweep

Script.

```
bash experiments/run_06_sparse_block_sweep.sh
```

Default sweep.

```
BLOCKS="3 5 7 9"
```

Configuration.

```
TSLRG_ANOMALY_MODE=anchored
TSLRG_RADIUS_MODE=anchor
TSLRG_RADIUS=1
TSLRG_PATCH_MASK_MODE=block
TSLRG_PATCH_MASK_RATIO=0.10
```

What it tests. Synthetic patches are selected in connected spatial blocks rather than independent random positions.

Reasoning. Many defects have spatial continuity: scratches, stains, dents, missing material, or texture corruption. Block masks are a crude feature-space analogue of image-level synthetic anomaly masks. They may give a localization head a more realistic spatial pattern than independent random patch corruption.

Expected behavior. Moderate block sizes should improve region-level metrics. Too-small blocks may look like random isolated noise; too-large blocks may again become global distribution shifts.

Interpretation. If block masks outperform random masks, spatial coherence is useful. If random masks outperform blocks, real defects may be more local or the block sizes may be too large for the chosen class.

4.8 Experiment 7: Gradient-Guided PCA Refinement

Script.

```
bash experiments/run_07_gradient_refinement_sweep.sh
```

Default sweep.

```
REFINE_STEPS_LIST="1 2 3"
REFINE_STEP_SIZES="0.02 0.05 0.1"
```

Configuration.

```
TSLRG_ANOMALY_MODE=anchored
TSLRG_RADIUS_MODE=anchor
TSLRG_RADIUS=1
TSLRG_PATCH_MASK_MODE=block
TSLRG_PATCH_MASK_RATIO=0.10
TSLRG_PATCH_MASK_BLOCK=5
```

What it tests. The generated fake features are refined using discriminator gradients before the main discriminator update. The update is projected back into the fitted patch PCA subspace and clamped by a Mahalanobis radius.

Reasoning. This experiment is motivated most directly by GLASS [8]. Randomly sampled negatives may be too easy or may miss the current decision boundary. Gradient refinement converts the generator into a hard-negative sampler: it searches for feature perturbations that the current discriminator should learn to reject, while still respecting the local PCA geometry.

Expected behavior. One or two refinement steps may improve both image and pixel metrics by tightening the decision boundary. Too many steps or too-large step sizes may create adversarial artifacts that no longer resemble plausible defects.

Interpretation. If refinement helps, then the generator benefits from discriminator feedback. If it hurts, the refinement may be producing adversarial directions that are useful for fooling the discriminator but not aligned with real anomalies. The next step would be to log before/after discriminator scores and feature norms.

5 Ablation Table Template

Experiment	Anchor	Radius	Mask	Refinement	Purpose
0	real	isotropic σ	all	no	SimpleNet baseline
1	real	threshold	all	no	previous anchored setup
2	real	fixed	all	no	magnitude decoupling
3	real	patch-calibrated	all	no	local scale calibration
4	real	anchor-gap	all	no	near-boundary negatives
5	real	anchor-gap	random	no	sparse local supervision
6	real	anchor-gap	block	no	coherent local supervision
7	real	anchor-gap	block	yes	hard negatives

6 How to Interpret Outcomes

6.1 Image AUROC Improves, Pixel Metrics Do Not

The discriminator has learned image-level evidence, but the fake samples are not teaching precise spatial localization. Prefer sparse masks, block masks, and smaller or anchor-calibrated radii.

6.2 Pixel Metrics Improve, Image AUROC Drops

The synthetic signal may be too subtle or too sparse for robust image-level ranking. Increase mask ratio, increase radius, or use a curriculum from stronger to subtler anomalies.

6.3 Training p_{fake} Is High, Test Metrics Are Weak

The fake anomalies are too easy or unrealistic. The discriminator is solving the synthetic task rather than learning real-defect evidence. Reduce radius, use anchor-gap calibration, or inspect PCA/residual energy.

6.4 Gradient Refinement Hurts

The refinement may be generating adversarial artifacts rather than plausible defects. Reduce `TSLRG_REFINE_STEP_SIZE`, reduce `TSLRG_REFINE_STEPS`, or set `TSLRG_REFINE_MAX_RADIUS`.

6.5 Sparse Blocks Beat Sparse Random Masks

Spatial coherence matters. This supports adding richer feature-space masks, such as lines, ellipses, texture-like blobs, or masks copied from image-level anomaly synthesis methods.

7 Recommended Execution Order

1. Run Experiment 0 and Experiment 1 on the same class. This establishes the vanilla-noise vs anchored-PCA comparison.
2. Run Experiment 2, then the patch and anchor radius sweeps. Stop early if one radius family is clearly unstable.
3. Take the best radius setting into sparse random and sparse block sweeps.
4. Run gradient refinement only after a good sparse-mask configuration is found. Refinement is more expensive and easier to misread without a stable non-refined baseline.
5. Repeat the winning configuration across additional classes.

For a quick smoke run:

```
CLASSNAME=screw META_EPOCHS=1 GAN_EPOCHS=1 \  
  bash experiments/run_01_anchored_threshold.sh
```

For the full single-class suite:

```
CLASSNAME=screw bash experiments/run_all_proposed_experiments.sh
```

8 Presentation Summary

- The baseline asks whether random feature noise is enough.
- Anchored PCA asks whether feature directions should follow the normal data manifold.
- Radius sweeps ask how close synthetic negatives should be to the normal boundary.
- Sparse masks ask whether localization requires local fake supervision.
- Gradient refinement asks whether the generator should adapt to the current discriminator.

The strongest possible result would show a monotonic story: random noise $\rightarrow$ anchored PCA improves image detection; calibrated radius improves localization; sparse block masks improve PRO; gradient refinement improves the hardest classes without destabilizing training.

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